NaviGate, ADP & e-NP

IT/ICT Department Reference Guide

1. Introduction

This document provides comprehensive network, firewall, and system requirements for three critical navigation software systems used on vessels: NaviGate, Admiralty Digital Publications (ADP), and electronic Nautical Publications (e-NP). IT/ICT departments should use this guide to ensure proper whitelisting and system configuration for both Master and Backup PCs.

⚠ **PLATFORM COMPATIBILITY:** These software applications are developed exclusively for Microsoft Windows. They are NOT natively developed for Apple macOS or iOS devices. Installation and operation on Apple devices is not supported.

⚠ **IMPORTANT:** IP addresses may change. Where possible, whitelist domain names/URLs instead of IP addresses for long-term reliability.

2. Critical Infrastructure Requirements

The following requirements are mandatory for proper operation of NaviGate, ADP, and e-NP software. Non-compliance may result in software malfunction, data corruption, or system instability.

2.1 Dedicated Hardware Requirements

> ⚠ **MANDATORY: DEDICATED PCs ONLY**
>
> PCs hosting NaviGate, ADP, and e-NP must be dedicated exclusively to navigation functions. These machines must NOT be used for general office tasks, email, web browsing, or any other purposes.

Hardware Configuration – Mandatory Requirements:

- **Dedicated Physical PC:** Each navigation software installation requires a dedicated physical computer for that sole purpose.
- **Network Connection:** Dedicated LAN connection (Ethernet) is **strongly recommended**. WiFi is acceptable but less reliable. The network should be dedicated or segmented for navigation systems.
- **Microsoft-Supported Windows:** The operating system must be a version of Windows that is currently supported by Microsoft with active security updates.
- **Full Administrator Rights:** The software must run with **full Administrator privileges**. Restricted user accounts will cause operational failures.

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Not Supported Configurations:

NOT SUPPORTED 100%	Reason
Thin Clients	Insufficient local processing power and storage; network dependency issues
Virtual Machines (VMs)	Performance unpredictability; resource contention; licensing concerns
Servers	Not designed for server environments; requires dedicated workstation
Shared/Multi-purpose PCs	Resource conflicts; potential data corruption; security risks

2.2 Cloud Synchronization Restrictions

⚠ WARNING: CLOUD SYNCHRONIZATION AGENTS

If cloud synchronization agents (DropBox, OneDrive, Google Drive, iCloud, or similar services) are installed on the navigation PCs, strict configuration rules must be followed to prevent data corruption and software malfunction.

Mandatory Sync Exclusion Rules:

- **Software Installation Folders:** All files related to NaviGate, ADP, and e-NP installations must be **completely excluded** from any synchronization.
- **Data/Archive Folders:** Navigation data archives and working files must NOT be synchronized to cloud storage.
- **Desktop Sync Prohibition:** Full desktop synchronization must be **disabled**. Syncing the entire desktop folder can cause file locking issues, version conflicts, and data corruption.
- **Documents Folder:** If any navigation software stores data in the Documents folder, that location must be excluded from sync.

Affected Services Include (but are not limited to):

Microsoft OneDrive, DropBox, Google Drive, iCloud Drive, Box, Nextcloud, Syncthing, and any other file synchronization software.

Best Practice: *If possible, do not install cloud synchronization agents on dedicated navigation PCs at all.*

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3. NaviGate Requirements

NaviGate is a software suite for Microsoft Windows consisting of three applications: NaviGate.exe (main program with GUI), nGateway.exe (communication gateway running as Windows service), and nGatewayManager.exe (administration tool for gateway settings).

3.1 Network / Firewall Configuration

The following IPs, ports, and URLs must be whitelisted for NaviGate to exchange data with the remote server:

Server	Protocol	Port	IP Address	Application
maritimegate.net	HTTPS	443	52.30.174.254	nGateway.exe
maritimegate.net	HTTP (fallback)	80	52.30.174.254	nGateway.exe

Note: HTTP on port 80 is not recommended but can be used if HTTPS is not supported by the vessel's network. To switch protocols, use the NG Gateway Admin Tool and replace the server address accordingly.

3.2 Windows System Requirements

Hardware and software requirements for NaviGate installation:

Component	Requirement
Operating System	Windows 7 SP1 (minimum) – Windows 10/11 recommended
Processor	Intel quad-core i5 or higher
RAM	8 GB minimum
Display Resolution	1360 × 768 pixels or higher
Disk Space	250 MB (applications) + 5 GB minimum (data archive)
.NET Framework	4.8 or higher (mandatory)
TLS Protocol	TLS 1.2 (must be enabled)
User Privileges	Full Administrator rights (mandatory)

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3.3 Windows Version-Specific Configuration

Windows 10/11 (Recommended)

For Windows 10 and Windows 11 systems, IT must verify the following before NaviGate installation:

- TLS 1.2 protocols are enabled and properly configured
- .NET Framework 4.8 or higher is installed

These operating systems provide native TLS support, better .NET Framework 4.8 compatibility, and receive continuous Microsoft security updates.

Windows 7 (Not Recommended)

⚠ **SECURITY WARNING:** Microsoft ended support for Windows 7 in January 2020. This operating system is vulnerable and no longer recommended for production environments.

Key issues with Windows 7:

- TLS 1.1/1.2 are NOT enabled by default
- Manual TLS enabling may not work properly
- Requires .NET Framework 4.8 with Service Pack 1 (mandatory)
- Requires KB3140245 update (and sometimes later rollups) for TLS 1.2 support

Installation Steps for Windows 7:

1. Verify Service Pack 1 is installed
2. Manually enable TLS 1.1 and TLS 1.2 in system settings (see Section 3.4)
3. Download .NET Framework 4.8 offline installer
4. Run installer as administrator
5. Install NaviGate using NaviGate_Setup.exe (requires Administrator rights)
6. Run NaviGate and proceed with activation

3.4 TLS 1.2 Configuration Guide

NaviGate and GateWay communicate via standard TCP connection and require TLS v1.2 to ensure security. Without TLS 1.2 properly enabled, communication will be completely dropped.

⚠ Important notes before starting:

- Windows 7 only supports TLS 1.2 if **Service Pack 1 (SP1)** is installed **and** the ***KB3140245*** update (and sometimes later rollups) has been applied. Without this, TLS 1.2 won't appear in Internet Options.
- Editing the registry incorrectly can cause system issues. Always **export the registry keys** before making changes (step well described in this guide).

Part A: Enable TLS 1.2 via Internet Options

1. **Open Internet Options**
   Press **Win + R**, type ***inetcpl.cpl*** and hit Enter.
2. **Go to Advanced Settings**
   In the Internet Properties window, click on the **Advanced** tab.
3. **Scroll to Security section**
   Near the bottom, you'll see checkboxes for SSL and TLS versions.
4. **Enable TLS 1.2**
   Check: ☑ **Use TLS 1.2**
   Uncheck older protocols if not needed (e.g., SSL 2.0, SSL 3.0, TLS 1.0) for better security.
   Click **OK** to save changes.

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Part B: Cleaning Custom TLS/SSL Settings in Registry

Windows stores custom overrides for TLS/SSL protocols under the **Protocols** key. If someone or some software forced protocol settings, you can delete them to fall back to Windows defaults.

Steps to Clean Up:

1. **Open Registry Editor**
   Press **Win + R**, type ***regedit*** and press Enter.
2. **Navigate to Protocols key**
   Browse to:
   HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols
3. **Backup before deleting**
   Right-click **Protocols** → **Export** → save a .reg backup file.
4. **Remove custom overrides**
   Delete all the subkeys under Protocols (**except TLS 1.2!**) if there are any.
   Example: Right-click on **SSL 2.0** → **Delete**. Repeat for TLS 1.0, TLS 1.1, etc. if present.
5. **Reboot Windows**
   Restart your PC so Schannel reloads with default settings.

✓ After this:

- Your **Internet Options** settings take effect normally.
- You can confirm TLS 1.2 works by visiting a site that only supports TLS 1.2, e.g. https://www.howsmyssl.com/

4. Admiralty Digital Publications (ADP) Requirements

ADP requires internet access for product activation and updates. The following network configuration must be whitelisted:

4.1 Network / Firewall Configuration

Parameter	Value
IP Address	137.135.243.254
Ports	80 and 443
Domain Name	www.admiralty.co.uk

4.2 URLs to Whitelist

- https://www.admiralty.co.uk/UPDATES/ADP/updating (Internet updates)
- https://adpactivation.ukho.gov.uk (Activation – uses IP 62.172.108.45, Ports 20-21)
- https://www.admiralty.co.uk/UPDATES/ADP/ (Website)
- sectigo.com (Certificate validation)

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5. e-Navigator Requirements

e-Navigator is used for managing UKHO digital products. The following network configuration is required:

5.1 Network / Firewall Configuration

Parameter	Value
IP Address	62.172.108.10
Ports	80 and 443
Domain Name	www.enavigator.ukho.gov.uk
URL	https://enavigator.ukho.gov.uk

6. Electronic Nautical Publications (e-NP) Requirements

e-NP within the Admiralty Gateway uses an FTP connection (PASV mode) with the normal control port of 21. Updates are processed through e-Navigator, and downloading of e-NPs is through the FTP site. Both addresses and URLs below must be whitelisted.

6.1 Network / Firewall Configuration

Parameter	Value
IP Addresses	62.172.108.10 and 137.135.243.254
Ports	80 and 443
FTP Port	21 (PASV mode)
Domain Name	www.enavigator.ukho.gov.uk

6.2 URLs to Whitelist

- https://enavigator.ukho.gov.uk
- ftp://ukho.gov.uk

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7. Quick Reference Summary

Complete whitelisting requirements for all three systems:

Software	IP Address(es)	Ports	Primary URL/Domain
NaviGate	52.30.174.254	443 (80)	maritimegate.net
ADP	137.135.243.254	80, 443	www.admiralty.co.uk
e-Navigator	62.172.108.10	80, 443	enavigator.ukho.gov.uk
e-NP	62.172.108.10, 137.135.243.254	80, 443, 21	enavigator.ukho.gov.uk, ftp://ukho.gov.uk

7.1 Windows Configuration Summary

Windows Version	Requirements	Recommendation
Windows 7	SP1 + manual TLS setup + .NET 4.8 offline installer	NOT RECOMMENDED
Windows 8+	Verify TLS 1.2 + .NET Framework 4.8+	RECOMMENDED

7.2 Infrastructure Requirements Summary

✓ REQUIRED	✗ NOT SUPPORTED 100%
Dedicated physical PC	Thin Clients
Dedicated LAN (preferred) or WiFi	Virtual Machines
Microsoft-supported Windows	Server installations
Full Administrator rights	Shared/multi-purpose PCs
Cloud sync exclusion for software files	Full desktop synchronization

8. Additional Notes

- **Platform Compatibility:** These applications are Windows-only. They are not natively developed for Apple macOS or iOS devices.
- **Master and Backup PCs:** Both machines must have internet access configured as per the whitelists above for navigational data download.
- **IP Address Changes:** IP addresses may change over time. Where possible, whitelist domain names/URLs rather than IP addresses for long-term reliability.
- **Security Recommendation:** Always use HTTPS (port 443) where available. HTTP (port 80) should only be used as a fallback if HTTPS is not supported by the vessel's network infrastructure.
- **Windows Upgrade:** While Windows 7 still works for NaviGate, upgrading to the latest version of Windows is strongly recommended for security reasons.
- **Administrator Rights:** NaviGate installation requires Administrator rights. Contact your IT department to obtain appropriate login credentials.

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